MS-CSSD-15
CSSD and Sterilization Practices

MODULE 10

Flexible Endoscope Reprocessing

Semi-critical, heat-labile, mechanically complex, and the source of the most serious documented outbreaks.

Learning hours	90 minutes
Who this module is for	Endoscopy nursing and technicians, CSSD staff, infection control
Assessment	Knowledge check plus an endoscope reprocessing observation
Standards basis	NABH 6th Edition IPC chapter; recognised endoscope reprocessing standards and manufacturer instructions

This module is one part of a certified course. Content is current as at August 2026; verify against the latest WHO, JCI and NABH publications and your own hospital policy before relying on it for a compliance decision.

Module 10 — Flexible Endoscope Reprocessing

Flexible endoscopes combine every difficulty in reprocessing: they are heat-labile so cannot be steam sterilised, they are mechanically complex with multiple narrow channels, they are heavily soiled in use, and they are expensive enough that hospitals own few and turn them round fast. They have caused several of the most serious documented reprocessing outbreaks.

What this module will enable you to do

- Describe the full reprocessing sequence and the purpose of each step.
- Explain why bedside pre-cleaning and leak testing cannot be omitted.
- Describe manual cleaning including channel brushing, and why it is irreplaceable.
- Explain the specific risk of the duodenoscope elevator mechanism.
- Describe drying and storage and why they matter.
- State the traceability and surveillance requirements for endoscopes.

10.1 Why endoscopes are different

- **Heat-labile:** cannot be steam sterilised, so high-level disinfection is the usual process, which has a smaller margin than sterilisation.
- **Multiple narrow channels** of different diameters, some of which are difficult to access and some of which are easy to forget.
- **Heavily contaminated in use** with organic material and a high bacterial load.
- **Biofilm forms readily** in channels, and once established it resists chemical disinfection entirely.
- **Expensive**, so hospitals own few and there is constant pressure on turnaround time.
- **Fragile**, so damage is common and a damaged scope admits fluid into the interior where it cannot be cleaned.
- **Complex and model-specific:** two scopes that look similar may have different channels, connectors and instructions.

10.2 The sequence

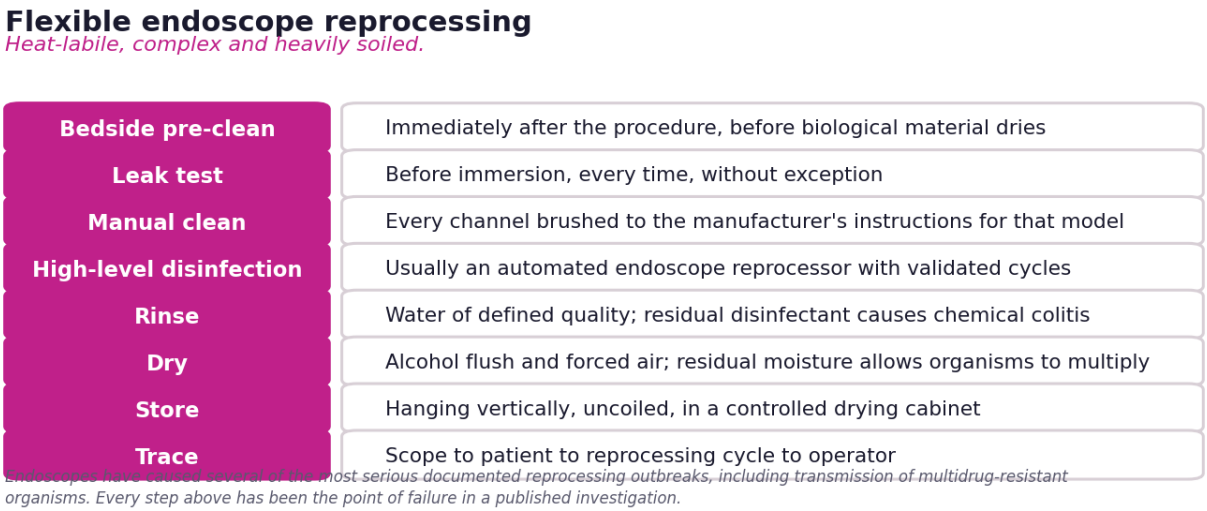


Figure 10.1 — The reprocessing sequence.

Bedside pre-clean

Immediately after the procedure, before the scope leaves the room. Wipe the insertion tube and suction or flush the channels with the agreed solution. Organic material begins to dry and to form biofilm within minutes, and once it has, no subsequent step reliably removes it. This is the step performed by the busiest person at the busiest moment and it is the one most often abbreviated.

Leak test

> **Before immersion, every time, without exception**
>
> A scope with a breach in its outer sheath or an internal channel that is immersed will admit fluid into its interior, where it cannot be cleaned or disinfected and where organisms will multiply. The scope is then contaminated internally and irreparably for reprocessing purposes, and it will be used on patients until the damage is found by other means. The leak test takes a minute and is skipped when the list is running late.

Manual cleaning

1. Fully immerse in the correct detergent at the correct dilution and temperature.
2. Wipe the exterior and the insertion tube.
3. **Brush every channel** to the manufacturer's instructions for that specific model, with a brush of the correct diameter and length, until the brush emerges visibly clean. Repeat until it does.
4. Brush the ports, valves, elevator recess and any other accessible cavity.
5. Flush all channels with detergent, then with water, using the connectors specified for that model.
6. Remove, clean and disinfect or replace all removable parts — valves, buttons, caps — which are frequently forgotten.
7. Inspect for damage and for residual soil before proceeding.

Brushing is the only step that removes biofilm

Chemical disinfection does not penetrate an established biofilm. Soaking does not remove it. Flushing does not remove it. Only mechanical action with a brush that physically contacts the channel wall disrupts it. This is why brushing is specified per channel, why brush diameter matters, and why omitting it has repeatedly caused outbreaks.

High-level disinfection

- Usually in an automated endoscope reprocessor with validated cycles, which provides a recorded, repeatable process.
- All channels connected to the correct connectors for that model; a channel not connected is not perfused and is not disinfected.
- The agent at the correct concentration, temperature and contact time; minimum effective concentration tested per protocol and the solution discarded at expiry regardless of test result.
- Manual high-level disinfection, where used, requires the same rigour and full immersion with all channels filled.
- Do not overload the reprocessor or process a scope alongside one it is not validated to be processed with.

Rinse, dry and store

- **Rinse** with water of defined quality; residual disinfectant causes chemical colitis and mucosal injury.
- **Dry** with an alcohol flush where protocol specifies, followed by forced air through every channel. Residual moisture allows waterborne organisms, particularly *Pseudomonas*, to multiply between uses.
- **Store** hanging vertically, uncoiled, with valves removed, in a controlled drying and storage cabinet.
- A defined maximum storage interval before reprocessing is required, per protocol.
- Transport between reprocessing and use in a manner that does not recontaminate.

10.3 Where it fails

Where endoscope reprocessing fails

Each of these has caused documented transmission.

The failure	The consequence
Delayed bedside pre-clean	Biological material dries in channels and forms biofilm
Leak test omitted	A damaged scope immersed contaminates the interior irreparably
Channels not brushed	Brushing is the only step that mechanically removes biofilm
Wrong connectors used	Channels not perfused during the automated cycle
Elevator mechanism not cleaned	The duodenoscope elevator is a documented outbreak site requiring specific attention
Inadequate drying	Residual moisture allows waterborne organisms to multiply between uses
Coiled storage	Fluid retained in channels
Rinse water contaminated	Recontaminates the scope at the final step
Manufacturer instructions not followed	Model-specific steps omitted in favour of a generic process
No traceability	Transmission cannot be investigated or contained

Figure 10.2 — The documented failure points.

Each of these has been identified in a published outbreak investigation. None is exotic.

1. **Delayed bedside pre-clean**, allowing material to dry and biofilm to establish.
2. **Leak test omitted**, and a damaged scope immersed.

3. **Channels not brushed**, or brushed with the wrong brush.
4. **Wrong connectors used** in the reprocessor, so a channel is not perfused.
5. **Elevator mechanism not cleaned** — the duodenoscope elevator is a documented outbreak site with a recess that is extremely difficult to clean and requires specific attention.
6. **Inadequate drying**, allowing organisms to multiply in residual moisture.
7. **Coiled storage**, retaining fluid in channels.
8. **Contaminated rinse water**, recontaminating the scope at the final step.
9. **Manufacturer instructions not followed**, a generic process applied to a model-specific device.
10. **No traceability**, so transmission cannot be investigated or contained.

The duodenoscope elevator

Duodenoscopes have an elevator mechanism at the distal tip which directs accessories. Its recess is narrow, mechanically complex and extremely difficult to clean, and it has been implicated in multiple international outbreaks of multidrug-resistant organisms in patients who had no other common exposure. Where your hospital uses duodenoscopes, the manufacturer's specific instructions for the elevator must be followed exactly, and any additional measures the manufacturer or your infection control team specify must be in place.

10.4 Surveillance and traceability

- **Traceability** linking scope to patient to reprocessing cycle to operator, for every procedure. Where a transmission is suspected, this is the only means of investigation.
- **Microbiological surveillance** of scopes at defined intervals per protocol, with defined actions on a positive result including quarantine and reprocessing.
- **Reprocessor validation** and periodic testing, including water quality at the final rinse.
- **Records** of every reprocessing cycle, with the scope identity, cycle result and operator.
- **Damage and repair log**, since a scope returning from repair must be reprocessed before use and repair history is relevant to investigation.
- **A defined response** to a suspected transmission: quarantine the scope, review records, involve infection control and microbiology, and consider patient notification.

10.5 Staffing and pressure

Endoscope reprocessing is performed under sustained time pressure because scopes are expensive, lists are long and turnaround determines throughput. That pressure is the underlying cause of most of the failures above, and it is not resolved by reminding technicians to be thorough.

- Adequate scope inventory for the list volume, so that reprocessing is not on the critical path for every case.
- Adequate reprocessing capacity and staffing, so that the cycle is not shortened to meet the list.
- Trained and competency-assessed staff, with training specific to each scope model in use.
- A defined right to hold a scope back from use where reprocessing was not completed properly, supported by named management.
- Records of reprocessing time and of any deviation, so that the pressure is visible rather than absorbed.

Where the pressure lands

When a list is running late and one scope is available, the person who decides whether the brushing step was adequate is a technician under direct pressure from a room full of waiting staff. That decision should not rest on individual courage. It requires written policy, adequate inventory and visible management backing, and where those are absent the shortcuts will happen whatever the training says.

10.6 Setting up an endoscopy reprocessing service

1. **Dedicated space** with separation of dirty and clean, and ventilation appropriate to the chemicals used.
2. **Adequate reprocessor capacity** for the list volume, with a contingency when one machine is down.
3. **Scope inventory** sufficient that reprocessing is not on the critical path for every case.
4. **Drying and storage cabinets** of adequate capacity, since storage is part of the process.
5. **Model-specific connectors, brushes and adapters** for every scope in the inventory, with spares.
6. **Manufacturer instructions** for every scope model, accessible at the point of work.
7. **Trained and competency-assessed staff**, with training specific to each model, and enough of them to cover leave and sickness.
8. **Water treatment** for the final rinse, monitored and tested.
9. **Traceability system** linking scope, patient, cycle and operator.
10. **Microbiological surveillance programme** with defined frequency and defined actions on a positive result.

10.7 Responding to a suspected transmission

1. Quarantine the implicated scope immediately; do not reprocess it, since reprocessing destroys evidence.
2. Notify infection control and microbiology without delay.
3. Retrieve the traceability records: which patients were exposed to that scope, in what order, and what the reprocessing record shows for each episode.
4. Review reprocessing practice and records for the period, including reprocessor cycle records and any deviation.
5. Sample the scope and the reprocessor per microbiology advice, guided by a hypothesis rather than broadly.
6. Review the scope's damage and repair history.
7. Consider patient notification and screening, under the hospital's incident and disclosure policy and with senior clinical input.
8. Report externally where required.
9. Write it up, including what the records could and could not establish, since the latter is usually the most useful finding.

Do not reprocess the scope

The instinct on suspecting a problem is to clean the scope thoroughly. That destroys the evidence that would establish whether it was contaminated and with what. Quarantine it untouched, secure it, and let microbiology and infection control direct what happens next.

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Check your understanding

Answer before reading the responses. These mirror the style of the course assessment.

Q1. Why is the leak test performed before immersion, every time, without exception?

Answer: Because a scope with a breach in its outer sheath or an internal channel will admit fluid into its interior when immersed. That interior cannot be cleaned or disinfected, organisms multiply inside it, and the scope is then irreparably contaminated for reprocessing purposes while continuing to be used on patients until the damage is found by other means. The test takes about a minute and is skipped when the list is running late.

Q2. Why can chemical disinfection not substitute for channel brushing?

Answer: Because biofilm — organisms embedded in a protective matrix adhering to the channel wall — resists chemical disinfection entirely. Soaking does not remove it and flushing does not remove it. Only mechanical action with a brush that physically contacts the channel wall disrupts it. This is why brushing is specified per channel with a brush of defined diameter, and why omitting it has repeatedly caused outbreaks.

Q3. What is the duodenoscope elevator and why does it matter?

Answer: A mechanism at the distal tip of a duodenoscope that directs accessories. Its recess is narrow, mechanically complex and extremely difficult to clean, and it has been implicated in multiple international

outbreaks of multidrug-resistant organisms in patients with no other common exposure. Where duodenoscopes are used, the manufacturer's specific elevator instructions must be followed exactly, along with any additional measures the manufacturer or infection control specify.

Q4. Why must a reprocessed endoscope be dried thoroughly and stored hanging vertically and uncoiled?

Answer: Because residual moisture in channels allows waterborne organisms, particularly *Pseudomonas* and other Gram-negatives, to multiply between uses, so a correctly disinfected scope becomes contaminated in storage. A coiled scope retains fluid in its channels. Hanging vertically and uncoiled with valves removed, in a controlled drying cabinet, allows channels to drain and dry.

Q5. A list is running late, one scope is available, and a technician is unsure whether the brushing step was adequate. What should determine the outcome?

Answer: Written policy and a defined right to hold the scope back, supported by named management — not the individual courage of a technician under direct pressure from a room of waiting staff. The underlying causes are inadequate scope inventory and reprocessing capacity, which put reprocessing on the critical path for every case. Those should be escalated with the patient risk stated, and reprocessing time and deviations recorded so the pressure is visible rather than absorbed.

Q6. What must be traceable for every endoscopic procedure and why?

Answer: The scope identity, the patient, the reprocessing cycle and the operator. Where a transmission is suspected, this is the only means of investigating it: identifying which patients were exposed to a particular scope, whether reprocessing was completed, and by whom. Without it, a suspected transmission cannot be investigated or contained, and patient notification becomes impossible to scope.